STATISTICIAN VALIDATION PACKAGE

Statistical Methodology, Instruments, Decision Rules, and Computed Results (N = 50) for Expert Review

Doomscrolling Detection and Digital Mindfulness Mobile Application for Short-Form Video Platforms Using VADER and Fuzzy Logic

(Heuristic Risk-State Estimation System)

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1. Purpose of This Document

This package compiles every element of the study that bears on statistical validation, so that a statistician can review the planned (and, once data are available, the executed) analysis without needing to read the full thesis manuscript. It consolidates the research questions, study design, variables, instruments, data structure, statistical treatment, and decision thresholds drawn from Chapter 3 (Methodology) of the manuscript.

The study is framed as a bounded software-engineering evaluation rather than a clinical trial or a fully powered efficacy study. The intended statistical claims are correspondingly modest: short-term observed differences, baseline convergent association, descriptive software-acceptability evaluation, and expert plausibility appraisal. The statistician is asked to confirm that the chosen methods, assumptions, exclusion rules, and interpretation language are appropriate for that scope and for the data actually collected.

What the statistician is asked to validate

- Whether each statistical test is appropriate for its research question, data type, and study design.
- Whether the sample size and effective (post-exclusion) sample sizes support the stated level of inference.
- Whether assumption checks and their fallbacks (parametric vs. non-parametric) are correctly specified.
- Whether the multiple-comparison control and effect-size reporting are appropriately scoped.
- Whether the composite-index (DSI) construction and the convergent-association approach are statistically sound.
- Whether interpretation and reporting language match what the analyses can actually support.

Sections 11–12 contain explicit validation questions and a checklist. Section 13 holds blank result tables for the survey and behavioral data, to be completed once collection is finished. Section 14 is the statistician's assessment and sign-off.

2. Study Overview

The system is a privacy-preserving Android application that performs on-device estimation of doomscrolling-related risk on short-form video platforms (TikTok, Facebook Reels, and Instagram Reels). It combines threshold-based heuristics, a VADER-compatible text-first sentiment pipeline, an on-device Vision-Language Model fallback (Moondream 0.5B) for no-text items, and interpretable fuzzy rule-based inference. All processing is local; “detection” refers to computational risk estimation from observable proxies, not clinical diagnosis.

Evaluation is reported under three separate headings, which the statistician should keep distinct because they carry different evidentiary weight:

- Operational feasibility — successful deployment completion and absence of critical system failures (descriptive).
- Software acceptability — ISO/IEC 25010, TAM, and SME results (descriptive + reliability).

- Estimator plausibility — baseline convergent association together with sentiment-reliable coverage (correlational, exploratory).

2.1 Research Type and Design

Element	Specification
Research type	Applied software-engineering study with a pilot-scale field evaluation
Design	Two-group field deployment: intervention group vs. logging-only control group
Allocation	Concealed permuted-block allocation list; initial 1:1 target (n = 25 per group)
Time structure	Week 1 baseline (prompts disabled, both arms) ⟶ Week 2 deployment (prompts enabled for intervention only)
Within-subject element	Week 1 vs. Week 2 comparison within the intervention group
Phases	Phase 1 recruitment/consent/allocation; Phase 2 baseline profile + setup; Phase 3 Week 1 logging; Phase 4 Week 2 deployment; Phase 5 post-usage survey + data retrieval

3. Research Questions, Objectives, and Analytic Mapping

The five research questions map onto distinct analytic strategies. Research Question 1 is design/development and is addressed descriptively, not by an inferential test.

#	Research question (abbreviated)	Analytic treatment
RQ1	What privacy-preserving architecture and estimation framework can support risk estimation, with a 2-input behavioral fallback for sentiment-unreliable sessions?	Descriptive (design & development sections). Not an inferential hypothesis test.
RQ2	What short-term Week 1⟶Week 2 changes occur in logged usage metrics and self-reported doomscrolling scores, between arms and within the intervention group?	Independent-samples t-test on change scores (between groups); paired-samples t-test (within intervention group). Effect sizes + 95% CIs; Holm-Bonferroni correction.
RQ3	What baseline convergent association exists between the Week 1 DSI and self-reported doomscrolling scores (participants with ≥ 3 sentiment-reliable Week 1 sessions)?	Bivariate correlation (Pearson r or Spearman ρ), component-wise and composite. Reported as plausibility check.
RQ4	How do users evaluate the system using ISO/IEC 25010 and TAM?	Descriptive summaries (mean, SD); SUS composite; Cronbach's α for internal consistency; pre-set favorable thresholds.
RQ5	How do subject-matter experts evaluate the system's design, privacy, heuristic logic, and intervention structure?	Descriptive (mean, SD) on a 5-point scale + qualitative synthesis. Expert plausibility appraisal, not formal CVI.

4. Respondents, Sampling, and Sample Size

Element	Specification
Target population	Filipino Android users aged 18+ who actively use at least one target short-form video platform
Sampling technique	Non-probability purposive-convenience sampling
Inclusion criteria	Age ≥ 18; Android 8.0+ device passing compatibility screening; active user of ≥ 1 target platform; consent to install and to usage-data collection
Target sample size	Minimum N = 50, split as evenly as possible (initial target n = 25 per group)
Sample size rationale	Supports a balanced pilot deployment and post-usage evaluation within an undergraduate thesis scope; explicitly not a fully powered efficacy sample
Effective sample sizes	Analyses that exclude sentiment-unreliable sessions or participants with insufficient baseline data report their effective n separately
Inference emphasis	Between-group estimates interpreted with emphasis on effect sizes and 95% confidence intervals alongside p-values
Selection bias note	Purposive-convenience sampling may overrepresent users already receptive to self-regulation tools, potentially affecting acceptance and compliance estimates

5. Variables and Operationalization

The risk-estimation inputs and the self-report/evaluation variables, with data source and measurement unit. Level-of-measurement labels are provided as an aid for the statistician's review; the manuscript states units and sources, and these levels follow from them.

Variable	Data source	Unit / scale	Level (for review)
Session Duration	Accessibility-derived session tracker	minutes	Continuous / ratio
Video Dwell Time	AccessibilityService (time between content transitions)	seconds	Continuous / ratio
Negative Sentiment Density (NSD)	VADER text-path + VLM no-text path + logger	% (0–100)	Continuous / ratio
Swipe Count (supporting)	AccessibilityService scroll-event stream	count	Count (supporting log only)
Risk Score (per session)	Fuzzy inference, Center-of-Gravity defuzzification	0–100	Continuous (composite)
Doomscroll Severity Index (DSI, per week)	Mean of sentiment-reliable session Risk Scores	0–100	Continuous (composite)
Doomscrolling Scale score	Short-form 4-item self-report (Items 1, 2, 10, 12)	Likert sum	Ordinal, treated as interval
ISO/IEC 25010 items	Post-usage survey (Likert)	5-point Likert	Ordinal, treated as interval
TAM (PU, PEOU)	Post-usage survey (Likert, 6 items each)	5-point Likert	Ordinal, treated as interval

Variable	Data source	Unit / scale	Level (for review)
SUS (Usability)	10-item System Usability Scale	0–100 composite	Continuous (composite)
SME ratings	Structured expert rubric (Likert)	5-point Likert	Ordinal, treated as interval
Prompt-response profile	Built-in logger (Week 2, intervention)	frequencies / %	Categorical (descriptive)

Composite-index construction (Doomscroll Severity Index)

- Per-week DSI is the mean of that week's sentiment-reliable session Risk Scores: $DSI_w = (1 / m_w) \times \Sigma RiskScore_s$, where m_w is the number of week-w sentiment-reliable (doomscrolling-evaluable) sessions.
- Sessions flagged Sentiment-Unreliable are excluded from DSI. Because DSI is a mean (not a cumulative sum), exclusions affect coverage more than scale.
- Each session Risk Score (0–100) is produced by fuzzy inference over Video Dwell Time, NSD, and Session Duration, with output category centers at 16.67 / 50.00 / 83.33 (Safe / Warning / Critical) and a fixed-prior scoring framework held constant across both study weeks.
- DSI coverage is reported by week and study arm: retained participants, retained sessions, and excluded sentiment-unreliable sessions.

6. Instruments

Instrument	Constructs / items	Response scale	Source / status
ISO/IEC 25010 software quality	Functional Suitability, Performance Efficiency, Usability, Reliability	5-point Likert	Usability via SUS (below); other three researcher-developed from ISO sub-characteristic definitions, refined by pre-deployment expert review
System Usability Scale (SUS)	10 standardized items (Usability sub-characteristic)	5-point, scored to 0–100 composite	Adopted verbatim; benchmark mean 68.05 (SD 14.05) per Hyzy et al. (2022)
TAM	Perceived Usefulness (6 items), Perceived Ease of Use (6 items)	5-point Likert	Contextualized to the doomscrolling estimation setting; administered in Phase 5
Short-form Doomscrolling Scale	4 items (Items 1, 2, 10, 12 of original), 7-day recall stem	Likert	Satici et al. (2023); Sharma et al. (2022). Administered end of Week 1 and end of Week 2
SME rubric	Technical soundness, input-range plausibility, 27-rule coherence,	5-point Likert + qualitative	Panel of 3 experts (software engineering; data

Instrument	Constructs / items	Response scale	Source / status
	architecture/privacy, intervention design, ISO 25010 overall		science/ML/fuzzy logic; digital well-being/psychology)

Instrument review and reliability safeguards.

- Before deployment, the combined survey undergoes structured expert content review for item relevance, clarity, redundancy, and construct alignment; unclear or overlapping items are revised.
- After collection, internal consistency for each multi-item sub-scale is estimated with Cronbach's α, interpreted with caution given the pilot sample size and the mix of standardized and adapted items.
- Open-ended responses are independently reviewed by two researchers, with coding differences reconciled through discussion; recurring issue categories are reported.

7. Data Structure Delivered for Analysis

For each participant, the dataset combines automated logs and self-report at the week level, keyed only by a participant study code (the name-to-code key is held separately). The statistician should expect the following analyzable structures:

Data block	Granularity	Key fields
Baseline profile	Per participant (Phase 2)	Study code, demographics, estimated short-form use, platform-use habits, study arm
Week 1 logs	Per session, aggregated to per-week	Session duration, video dwell time, NSD, swipe totals, reliability/sentiment-reliable flags, Week 1 DSI
Week 2 logs	Per session, aggregated to per-week	Same metrics as Week 1, plus prompt-response logs (intervention arm), Week 2 DSI
Self-report	Per participant × week	Short-form Doomscrolling Scale score (end of Week 1, end of Week 2)
Post-usage survey	Per participant (Phase 5)	ISO/IEC 25010 items, TAM PU/PEOU items, SUS items, open-ended feedback
SME evaluations	Per expert (n = 3)	Rubric Likert ratings + qualitative comments
Profiling records	Reference-device set + field traces	CPU, RAM, battery (descriptive supporting evidence)

Change-score derivation. For RQ2, the primary quantities are change scores (Week 2 minus Week 1) per participant for each behavioral outcome and for the self-report score. Two measurement variants of the time-based metrics are retained:

- Raw elapsed metrics — raw elapsed session duration and raw elapsed video dwell time (primary).

- Prompt-excluded active-use metrics — time inside the intervention is excluded until the participant returns to the platform (supplementary transparency check).

8. Statistical Treatment

This is the core section for validation. Each method below is reproduced as specified in the manuscript, organized by analytic purpose.

8.1 Descriptive Summaries (ISO/IEC 25010 & TAM Survey Data)

Likert responses from the ISO/IEC 25010 quality survey and the TAM instrument (PU and PEOU subscales) are summarized using mean scores and standard deviation. Means are described on a 5-point scale where 3.00 is the neutral midpoint.

8.2 Objective Acceptance Thresholds (Decision Aids)

Pre-set interpretation thresholds are established as decision aids for Chapter 4 to reduce post-hoc rationalization. They are study-defined favorable benchmarks, not universal cutoffs, and results below them are reported as unmet software-acceptability targets.

Measure	Favorable threshold	Basis
System Usability Scale (SUS)	Composite ≥ 70	Set slightly above the 68.05 digital-health benchmark mean (Hyzy et al., 2022)
ISO/IEC 25010 subscales (FS, PE, Reliability)	Mean ≥ 3.50 / 5	Study-defined favorable-evaluation target
TAM constructs (PU, PEOU)	Mean ≥ 3.50 / 5	Study-defined favorable-evaluation target
SME evaluation	Mean ≥ 4.00 / 5	Study-defined favorable-evaluation target for expert review

Objective CPU, RAM, and battery observations are summarized descriptively across deployment and controlled profiling. Given device/hardware and manufacturer background-process differences, these are interpreted as supporting evidence for Performance Efficiency rather than a uniform decision criterion.

8.3 Cronbach's Alpha (Instrument Reliability)

Cronbach's α is computed for each sub-scale (Functional Suitability, Performance Efficiency, Usability, Reliability, PU, PEOU). A pre-set threshold of $\alpha \geq 0.70$ is adopted as a conventional descriptive benchmark. Given the pilot sample and the mix of standardized and adapted items, α is treated as a reliability check rather than a stand-alone validation claim.

8.4 Primary Behavioral Analysis (RQ2)

Between-group (primary): Independent-samples t-test on change scores (Week 2 − Week 1), comparing intervention vs. control. Using change scores rather than raw Week 2 values adjusts for chance baseline imbalance and individual differences.

Within-group (secondary): Paired-samples t-test comparing Week 1 vs. Week 2 within the intervention group.

Given the pilot sample, these are treated as exploratory analyses estimating short-term observed differences under study conditions, not precise intervention-effect magnitudes. Separate primary tests are conducted for:

- Mean daily raw elapsed session duration (minutes)
- Mean daily raw elapsed video dwell time (seconds)
- Mean daily Negative Sentiment Density (%) — computed only from sentiment-reliable sessions; participant-weeks with no sentiment-reliable sessions contribute missing data for this outcome only, with analyzed n reported
- Pre/Post Doomscrolling Scale score (self-reported)

A supplementary transparency check repeats the session-duration and dwell-time tests on prompt-excluded active-use metrics; the difference between raw-elapsed and prompt-excluded results is reported to show the measurement-rule effect.

8.4.1 Multiple-Comparison Control and Effect Sizes

To control family-wise Type I error across the four change-score comparisons, p-values are adjusted with the Holm-Bonferroni sequential correction at an initial family-wise $\alpha = 0.05$. For each comparison, Cohen's d and its 95% confidence interval are reported alongside the Holm-adjusted p-value, to support effect-size interpretation rather than significance alone.

8.4.2 Handling of Assumption Violations

Test	Assumption check	Fallback if violated
Independent-samples t-test (change scores)	Normality of change scores within each group (Shapiro-Wilk); equal variances (Levene's test)	Mann-Whitney U if normality severely violated; Welch's t-test if variances unequal but normality holds
Paired-samples t-test (within intervention)	Normality of within-subject differences (Shapiro-Wilk)	Wilcoxon Signed-Rank Test
Any comparison after exclusions	Stable analyzable sample size	If too small for stable inference, reported descriptively with effective n and directional pattern — not overstated as confirmatory

8.5 Descriptive Analysis of Prompt-Response Logs

Week 2 prompt-response data (e.g., continue, dismiss, take-break) are summarized using frequency counts and percentages. These logs are descriptive evidence of intervention interaction only and are not analyzed as a baseline-comparable outcome variable.

8.6 Baseline Convergent Association Analysis (RQ3)

Assessed on Week 1 data only, in two steps:

1. Component-wise correlation table — bivariate association between each individual input (mean daily session duration, mean daily video dwell time, mean daily NSD) and the baseline Doomscrolling Scale score.
2. Composite correlation — bivariate correlation between the full composite DSI (fixed priors) and the Doomscrolling Scale score.

The pattern of component-wise vs. composite correlations is discussed descriptively to assess whether the multi-variable composite associates more strongly with self-reported doomscrolling than session duration alone. Pearson's r is used when linearity and normality hold; otherwise Spearman's ρ. Results are interpreted as a baseline plausibility check, not diagnostic validation.

Inclusion rule: only participants with ≥ 3 sentiment-reliable Week 1 sessions ($m_1 \geq 3$) are included; the effective sample size (N_effective) is reported. The ≥ 3 rule is a pragmatic weekly floor for forming the baseline aggregate (Buekers et al., 2025; Meyer et al., 2022; Ratitch et al., 2023; Yao et al., 2021).

Why supervised classification metrics are NOT used

- Traditional supervised metrics (precision, recall, F1, AUROC) are not estimated, because session-level ground-truth labeling would require interrupting active scrolling.
- Convergent association is therefore the main plausibility check, and the system is interpreted as a heuristic behavioral risk estimator rather than a validated classifier.

8.7 Significance Level and Interpretation

For these exploratory pilot analyses, the nominal significance level is $\alpha = 0.05$. Statistically significant Week 1–Week 2 differences are interpreted as observed short-term behavioral differences under study conditions. The two-group randomized design supports stronger inference than a one-group pre/post comparison, but findings are not interpreted as long-term intervention effects or calibrated validation. Prompt-excluded active-use metrics are interpreted separately because they incorporate the prompt-display rule.

8.8 Descriptive Coding of Open-Ended Feedback (Phase 5)

Open-ended responses are analyzed through light inductive coding: two researchers read responses in full, group them into recurring issue categories, and reconcile differences through discussion before summarizing. The goal is to surface usability issues, perceived intervention value, and implementation constraints not captured by the Likert measures.

8.9 Missing Data and Exclusion Criteria

- Convergent correlation: participants need $m_1 \geq 3$ sentiment-reliable Week 1 sessions for Week 1 DSI to enter the dataset; the edge case $m_1 = 0$ leaves DSI undefined by construction. N_effective after exclusion is reported.

- NSD outcome: participants require ≥ 1 sentiment-reliable day in each compared week to contribute a valid week-level comparison; otherwise that outcome is treated as missing for that participant.
- Paired pre/post analyses: complete paired observations are required per outcome; the analyzed n for each test is reported.

9. Decision Thresholds at a Glance

Domain	Statistic	Threshold / rule
Usability	SUS composite	≥ 70 (favorable)
ISO 25010 subscales	Mean (5-pt)	≥ 3.50 (favorable)
TAM (PU, PEOU)	Mean (5-pt)	≥ 3.50 (favorable)
SME ratings	Mean (5-pt)	≥ 4.00 (favorable)
Internal consistency	Cronbach's α	≥ 0.70 (descriptive benchmark)
Behavioral inference	α (family-wise)	0.05, Holm-Bonferroni across 4 change-score tests
Effect size	Cohen's d	Reported with 95% CI for each comparison
Convergent association	Pearson r / Spearman ρ	Pearson if normality + linearity hold; otherwise Spearman
Convergent inclusion	Sentiment-reliable Week 1 sessions	$m_1 \geq 3$ per participant

10. Points the Authors Flag for the Statistician's Attention

These are stated in the manuscript and surfaced here so the statistician can confirm the framing is defensible:

- Scope of inference — the study is pilot-sized (target N = 50) and explicitly not powered for efficacy; the authors lean on effect sizes and 95% CIs rather than pass/fail significance.
- Ordinal Likert data treated as interval — means/SDs and t-tests are applied to Likert-based subscales; the statistician may wish to confirm this is acceptable for the reporting claims, or recommend non-parametric reporting.
- Composite-then-correlate — DSI is a fuzzy-inference composite that is then correlated with self-report; the statistician should confirm the convergent-association interpretation does not overstate validation.
- Exclusion-driven sample reduction — sentiment-unreliable sessions reduce effective n on NSD and DSI outcomes; effective n is reported per outcome.
- Multiplicity scope — Holm-Bonferroni is applied across the four primary change-score comparisons; the statistician may confirm whether secondary/within-subject and prompt-excluded tests should share the family.

- Self-selection — purposive-convenience sampling may bias acceptance/compliance estimates upward.

11. Validation Questions for the Statistician

Please assess each item and record your response in Section 14 (or inline).

Q1. Are the independent-samples and paired-samples t-tests on change scores appropriate for this two-group, two-week design and for the distribution of the behavioral outcomes?

Q2. Is the Shapiro-Wilk / Levene's → Mann-Whitney U / Welch / Wilcoxon fallback ladder correctly specified and ordered?

Q3. Is Holm-Bonferroni the appropriate multiplicity control here, and is the family of four change-score comparisons scoped correctly? Should within-subject or prompt-excluded tests be included or kept separate?

Q4. Is Cohen's d (with 95% CI) the right effect-size choice for these comparisons, and is the CI method appropriate at this sample size?

Q5. Is treating 5-point Likert subscales as interval (means, SD, t-tests, Cronbach's α) defensible for the stated claims, or should non-parametric / ordinal methods be recommended?

Q6. Is Cronbach's $\alpha \geq 0.70$ a reasonable descriptive benchmark given the pilot n and the mix of standardized (SUS) and researcher-developed items? Should ω or other indices be considered?

Q7. Is the SUS scored and benchmarked correctly (0–100 composite; ≥ 70 target vs. the 68.05 benchmark)?

Q8. Is the Pearson-vs-Spearman decision rule for convergent association appropriate, and is the $m_1 \geq 3$ inclusion floor statistically reasonable?

Q9. Is the composite-then-correlate DSI approach sound, and is the component-wise vs. composite comparison interpreted without overstatement?

Q10. Are the missing-data and exclusion rules (per-outcome) handled in a way that avoids bias, and is reporting effective n per test sufficient?

Q11. Given target N = 50 (and likely smaller effective n after exclusions), is the interpretation language ('exploratory', 'observed short-term differences', 'plausibility check') appropriately calibrated?

Q12. Are there analyses the authors should add, drop, or reframe before Chapter 4 is finalized?

12. Statistician Validation Checklist

Mark each item Adequate / Needs revision / Not applicable, with a brief note.

Item	Assessment	Notes
Research questions correctly mapped to analyses		
Study design adequate for the inferential claims		

Item	Assessment	Notes
Sampling and sample-size rationale appropriate for scope		
Variables correctly operationalized and leveled		
Instruments and scoring (incl. SUS, Doomscrolling Scale) correct		
Change-score t-test approach appropriate		
Assumption checks and fallbacks correctly specified		
Multiplicity correction (Holm-Bonferroni) appropriately scoped		
Effect sizes and 95% CIs correctly reported		
Cronbach's α application and threshold reasonable		
Convergent-association method and inclusion rule sound		
Composite DSI construction statistically defensible		
Missing-data / exclusion handling unbiased		
Interpretation language calibrated to evidence		

13. Survey & Behavioral Results

Results below were computed from the final analysis workbook (50 respondents; 10,134 logged sessions, 87.0% sentiment-reliable). Survey descriptives, Cronbach's α, the SUS composite, and the RQ3 correlations reproduce the workbook's pre-computed values; the behavioral effect sizes, confidence intervals, and Holm-Bonferroni adjustment were computed here per the Section 8 specification, and the test for each outcome was re-selected by the methodology's decision rule after checking Shapiro-Wilk and Levene's. The workbook applied Welch's t-test uniformly to the change scores; this review found duration normal with unequal variances (Welch correct) but dwell, NSD, and Doomscrolling change scores non-normal, so those three are reported with the pre-specified Mann-Whitney U fallback, with the workbook's Welch p cross-referenced. Both routes reach identical conclusions. For NSD, the two intervention participants with no sentiment-reliable sessions are excluded under the coverage rule (effective n = 23). A negative effect size denotes a larger reduction in the intervention group (favorable).

13.1 Sample Disposition and Coverage

Metric	Intervention	Control	Total
Enrolled / allocated	25	25	50
Completed Week 1	25	25	50
Completed Week 2	25	25	50
RQ3-eligible ($m_1 \geq 3$ reliable Week 1 sessions)	23	25	48
Excluded — insufficient reliable sessions	2	0	2
Sessions logged (both weeks)	—	—	10,134

Session reliability: 8,817 reliable (87.0%); 1,317 sentiment-unreliable (13.0%). Platform mix: Instagram 3,433; Facebook 3,389; TikTok 3,312. Coverage events: HIGH_OOV 924; VLM_UNRESOLVED 241; EXTRACTION_FAILURE 152. Reported as descriptive context; the unreliable sessions account for the two coverage exclusions.

13.2 ISO/IEC 25010 — Descriptive Summary

Subscale	n	Mean	SD	Cronbach's α	≥ 3.50?
Functional Suitability	50	3.93	0.50	0.852 (Good)	Yes
Performance Efficiency	50	4.09	0.55	0.859 (Good)	Yes
Reliability	50	4.05	0.50	0.717 (Acceptable)	Yes

Usability is reported via SUS in 13.3. All three subscales meet the ≥ 3.50 favorable target; all α ≥ 0.70.

13.3 System Usability Scale (SUS)

Metric	Value
n	50
SUS composite (0–100)	80.95

Metric	Value
SD (per-respondent SUS)	14.83
Cronbach's α	0.957 (Excellent)
≥ 70 target met?	Yes (vs. 68.05 benchmark)

13.4 TAM — Descriptive Summary

Construct	n	Mean	SD	Cronbach's α	≥ 3.50?
Perceived Usefulness	50	4.12	0.40	0.750 (Acceptable)	Yes
Perceived Ease of Use	50	4.09	0.53	0.855 (Good)	Yes

13.5 SME Evaluation (RQ5)

Not yet collected

- The final workbook's SME sheets (sme_evaluation, sme_open_ended_feedback) contain headers only — no expert responses.
- RQ5 requires the completed forms from the 3-expert panel (5-point rubric; favorable target mean ≥ 4.00). This section will be filled when those data arrive.

13.6 Primary Behavioral Comparisons (Between Groups, Change Scores)

Outcome	n (I / C)	Test	Stat	p (Holm)	Cohen's d [95% CI]
Session duration (min)	25 / 25	Welch's t-test	−4.17	< .001	−1.18 [−1.78, −0.58]
Video dwell time (s)	25 / 25	Mann-Whitney U	124.0	< .001	−1.33 [−1.94, −0.72]
NSD (%) †	23 / 25	Mann-Whitney U	68.0	< .001	−1.82 [−2.50, −1.15]
Doomscrolling Scale	25 / 25	Mann-Whitney U	105.0	< .001	−1.50 [−2.13, −0.87]

† NSD excludes the two zero-coverage intervention participants (effective n = 23). Test choice follows the methodology's assumption-check rule: duration was normal with unequal variances (Welch); dwell, NSD, and Doomscrolling were non-normal (Shapiro-Wilk), so Mann-Whitney U was used. Workbook Welch p-values for cross-reference: duration .00024, dwell .00003, NSD .0000014, Doomscrolling .0000044 — all agree with the conclusions above. Holm-Bonferroni applied across the four primary outcomes at family-wise α = 0.05; all four favor the intervention.

13.7 Supplementary Outcomes (Sessions/Day and DSI)

Outcome	n (I / C)	Test	Stat	p	Cohen's d [95% CI]
Mean sessions/day	25 / 25	Mann-Whitney U	153.5	.002	−0.82 [−1.40, −0.24]
DSI (composite risk)	23 / 25	Mann-Whitney U	81.0	< .001	−1.84 [−2.52, −1.16]

These two outcomes are reported as supplementary in the workbook and are not part of the Holm family of four primary outcomes. Both favor the intervention. Note: the prompt-excluded active-use comparison specified in Section 8.4 is not separately computed in this export, although the raw and prompt-excluded durations are present in the session logs and can be added if the panel wants it.

13.8 Within-Group (Intervention) Paired Comparisons

Outcome	n	Test	Stat	p	Cohen's d_z [95% CI]
Session duration (min)	25	Paired t-test	t = −4.41	< .001	−0.88 [−1.34, −0.42]
Video dwell time (s)	25	Paired t-test	t = −5.45	< .001	−1.09 [−1.58, −0.59]
NSD (%) †	23	Wilcoxon signed-rank	W = 16.0	< .001	−1.28 [−1.83, −0.73]
Doomscrolling Scale	25	Wilcoxon signed-rank	W = 7.0	< .001	−1.25 [−1.78, −0.73]

† NSD effective n = 23. Within-subject differences for duration and dwell were normal (paired t-test); NSD and Doomscrolling were non-normal, so the Wilcoxon signed-rank fallback applied. d_z is the standardized within-subject effect.

13.9 Baseline Convergent Association (Week 1, N = 48)

Predictor (vs. Week 1 Doomscrolling Scale)	Spearman ρ (primary)	Pearson r	p
Mean daily session duration	0.92	0.82	< .001
Mean daily video dwell time	0.75	0.82	< .001
Mean daily NSD	0.76	0.83	< .001
Composite DSI	0.76	0.82	< .001

Week 1 DSI and mean daily duration depart from normality (Shapiro-Wilk $p < .001$) while the Doomscrolling Scale is approximately normal, so Spearman's ρ is the methodology-selected statistic; Pearson r is reported for reference. All associations are strong, positive, and significant. The composite DSI ($\rho = 0.76$) does not associate more strongly with self-report than its components — mean daily session duration shows a notably higher rank correlation ($\rho = 0.92$). This answers RQ3's design question directly: the DSI is convergent with self-report but does not outperform single behavioral inputs in this sample.

13.10 Prompt-Response Logs (Week 2, Intervention) — Descriptive

Logged event / response	Frequency	Percentage
TAKE_BREAK	191	51.6%
SHOWN	68	18.4%
SUPPRESSED (cooldown)	49	13.2%
CONTINUE	45	12.2%
DISMISSED	17	4.6%
VIEW_DASHBOARD	0	0.0%
Total prompt events	370	100%

Reported as logged. SHOWN and SUPPRESSED are engine states (prompt displayed; prompt withheld during cooldown); TAKE_BREAK, CONTINUE, and DISMISSED are user responses. Descriptive only — not a baseline-comparable outcome.

13.11 Open-Ended Feedback — Recurring Categories

Not yet collected

- The final workbook's open_ended_feedback sheet contains headers only — no participant responses.
- The inductive coding (recurring categories with frequencies) will be completed when the Phase 5 qualitative responses are provided.

13.12 Calibrated Interpretation Summary

Across all four primary outcomes, the intervention group showed larger short-term Week 1→Week 2 reductions than the logging-only control, all significant after Holm-Bonferroni adjustment, with large effect sizes ($|d| \approx 1.2–1.8$). The within-intervention paired analysis shows the same pattern. Software-acceptability targets were met across the board (6 of 6): all ISO/IEC 25010 subscales ≥ 3.50, SUS = 80.95 (above the ≥ 70 target and the 68.05 benchmark), both TAM constructs ≥ 3.50, and all reliability coefficients ≥ 0.70. Baseline convergent association is strong (composite DSI $\rho = 0.76$).

Consistent with the methodology's stated scope, these results are interpreted as observed short-term behavioral differences under study conditions and a baseline plausibility check — not as evidence of long-term efficacy or as calibrated estimator validation. The pilot N = 50 (effective n = 48 for RQ3; 23/25 for NSD) supports exploratory, effect-size-led interpretation rather than confirmatory claims. RQ5 (SME) and the open-ended qualitative analysis remain outstanding.

Three points the authors should be ready to discuss with the statistician

- Test selection: the workbook applied Welch's t-test to all change scores, but three of the four primary outcomes are non-normal, where the pre-registered rule specifies Mann-Whitney U. Conclusions are unchanged, but the reported test should match the assumption-check result per outcome.
- Effect sizes are large and uniform for a two-week behavioral pilot ($|d| \approx 1.2–1.8$ across distinct outcomes, all favoring the intervention). A reviewer will likely probe measurement independence and the logging/data-generation pipeline; be ready to walk through it.
- Convergent association does not favor the composite: Week 1 session duration alone has a higher rank correlation with self-report ($\rho = 0.92$) than the composite DSI ($\rho = 0.76$). Frame this honestly in Chapter 4 rather than claiming composite superiority.

14. Statistician's Assessment and Sign-Off

Overall assessment of the statistical methodology and (where applicable) the executed analysis:

Recommendation:

- ☐ Methodology is statistically sound as specified
- ☐ Sound with minor revisions (noted above)
- ☐ Requires substantive revision before analysis / Chapter 4

Statistician (printed name): ________________________________

Affiliation / credentials: ________________________________

Signature: ________________________________

Date: ________________________________

Source: Chapter 3 (Methodology), Sections 3.6–3.11, of the thesis manuscript "Doomscrolling Detection and Digital Mindfulness Mobile Application for Short-Form Video Platforms Using VADER and Fuzzy Logic." Section 13 results computed from the final analysis workbook (REINVENT_ANALYSIS.xlsx; N = 50): survey descriptives, Cronbach's α, SUS, and the RQ3 correlations reproduce the workbook's values, while the inferential tests, effect sizes, confidence intervals, Holm-Bonferroni adjustment, and per-outcome test selection were computed per the Section 8 specification. SME (RQ5) and open-ended feedback are not yet collected. Validation questions, checklist, and the sign-off block are review scaffolding.